

An exact arithmetic phase diagram for two-ray Carleson systems

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Status. Candidate exact classification theorem for a natural two-ray model; current verdict: likely valid. It includes sharpness of the source density theorem, but not the source paper's unrestricted spanning-properties program.

1 Source program and target

Krishtal and Miller [1] study the spanning properties of canonical Carleson systems. Their introduction says that a full understanding remains far from achieved and proposes a future sampling/interpolation program. The exact source passage appears below.

Perturbation results by themselves, however, seem to be insufficient to establish the conjecture from [9], let alone to fully understand the spanning properties of Carleson systems. In Section 2, we have accomplished the former goal by combining perturbation techniques with the celebrated Müntz-Szász theorem on the completeness of monomial systems in $C[0, 1]$. The latter goal is far from being achieved, although, we have made some headway in Section 3. We intend to pursue it in our future research, perhaps, utilizing techniques from sampling and interpolation in spaces of holomorphic functions [11, 13, 14] (or more general reproducing kernel Hilbert spaces [10]), more advanced perturbation techniques (such as in [12]), and the spectral theory of totally positive matrices [15].

For a countable set $\Lambda \subset [0, \infty)$, the paper defines its *logarithmic block density* by

$$L(\Lambda) = \inf_{\mu > 1} \frac{1}{\log \mu} \limsup_{t \rightarrow \infty} \sum_{\lambda \in \Lambda \cap [t, \mu t]} \frac{1}{\lambda}.$$

Its Theorem 3.5 states that if a Carleson spectrum lies in

$$\mathbb{D}_c = \{re^{i\phi} \in \mathbb{D} \setminus \{0\} : |\phi| \leq c\}, \quad 0 \leq c < \pi,$$

then $L(\Lambda) > c/\pi$ guarantees completeness of the canonical system $\{D^\lambda g\}_{\lambda \in \Lambda}$ in $\ell^2(\mathbb{N}_0)$.

Lemma 3.2 is all we need to establish the following completeness result.

Theorem 3.5. *Assume that $\mathcal{Z} \subset \mathbb{D}_c$ is a Carleson sequence, where*

$$\mathbb{D}_c = \{z = re^{i\theta} \in \mathbb{D} \setminus \{0\} : \theta \in [-c, c]\}, \quad c \in [0, \pi).$$

Assume also that $\Lambda \subset [0, \infty)$ satisfies $L(\Lambda) > \frac{c}{\pi}$. Then the system $\{D^\lambda g\}_{\lambda \in \Lambda}$ is complete in $\ell^2(\mathbb{N}_0)$.

The result below completely classifies every arithmetic fractional-power orbit for an explicit two-ray family, and then answers the natural sharpness question as a corollary.

2 Definitions

The pseudohyperbolic distance in $\mathbb{D}$ is

$$\rho(z, w) = \left| \frac{z - w}{1 - w\bar{z}} \right|.$$

A sequence $\mathcal{Z} = \{z_j\} \subset \mathbb{D}$ is a *Carleson sequence* when

$$\inf_k \prod_{j \neq k} \rho(z_k, z_j) > 0.$$

For such a spectrum, let D be diagonal on the coordinate basis, $D\delta_j = z_j\delta_j$, and let

$$g_j = \sqrt{1 - |z_j|^2}.$$

Following [1], if $z_j = r_j e^{i\phi_j}$, with $\phi_j \in [-\pi, \pi)$, fractional powers use the principal phase:

$$D^\lambda \delta_j = z_j^\lambda \delta_j = e^{\lambda(\log r_j + i\phi_j)} \delta_j.$$

3 Proof intuition

Put spectral points at hyperbolically geometric radii on the two rays $\pm\theta$. Sampling the orbit at times αk replaces the spectrum by its principal α -th powers. There are exactly two possibilities. If $\alpha\theta \in \pi\mathbb{Z}$, the two members of every radial pair collide after powering, so their coordinate differences give infinitely many annihilators. Otherwise the powered points lie on two distinct rays and still form a Carleson sequence. The sampled orbit is then a bounded invertible diagonal image of the canonical frame for that powered spectrum.

The key geometric fact is stable under every radial power. In hyperbolic coordinates $u = 2 \operatorname{artanh} r$, the map $r \mapsto r^\alpha$ has derivative bounded below along the chosen radial sequence. Thus same-ray products retain a uniform lower bound. Across the rays, distinct boundary limit points again give a summable, uniformly controlled loss. This yields an exact frame-versus-infinite-defect dichotomy, not only a counterexample at the critical density.

4 Exact arithmetic phase diagram

Theorem 1. *Fix $A > 0$ and $0 < \theta < \pi$, and interlace the two-ray spectrum*

$$r_n = \tanh(A(n+1)), \quad z_n^\pm = r_n e^{\pm i\theta} \quad (n \in \mathbb{N}_0).$$

This is a Carleson sequence. Let D be the corresponding diagonal operator and $g_n^\pm = \sqrt{1 - r_n^2}$. Then, for every $\alpha > 0$,

$$\{D^{\alpha k} g\}_{k \in \mathbb{N}_0} \begin{cases} \text{is a frame for } \ell^2(\mathbb{N}_0), & \alpha\theta \notin \pi\mathbb{Z}, \\ \text{has infinite-dimensional orthogonal complement,} & \alpha\theta \in \pi\mathbb{Z}. \end{cases}$$

Proof. We first establish a powered-spectrum claim. Let $\beta > 0$ satisfy $\beta\theta \notin \pi\mathbb{Z}$, and set

$$w_n^\pm = (z_n^\pm)^\beta = r_n^\beta e^{\pm i\beta\theta}.$$

Choose $\phi_\beta \in (-\pi, \pi)$ with $e^{i\phi_\beta} = e^{i\beta\theta}$. The nonresonance assumption gives $0 < |\phi_\beta| < \pi$, and the powered spectrum is the union of the distinct rays

$$w_n^\pm = s_n e^{\pm i\phi_\beta}, \quad s_n = r_n^\beta.$$

To control a single ray, introduce its hyperbolic radial coordinate

$$U_\beta(x) = 2 \operatorname{artanh}(\tanh(x)^\beta), \quad x \geq A.$$

Its derivative is

$$U'_\beta(x) = \frac{2\beta \tanh(x)^{\beta-1} \operatorname{sech}^2 x}{1 - \tanh(x)^{2\beta}} > 0.$$

Moreover $U'_\beta(x) \rightarrow 2$ as $x \rightarrow \infty$. After adjoining the point at infinity, this is a positive continuous function on the compact interval $[A, \infty]$. Hence

$$a_\beta := \inf_{x \geq A} U'_\beta(x) > 0.$$

Writing $x_n = A(n+1)$, the mean-value theorem yields

$$|U_\beta(x_m) - U_\beta(x_n)| \geq a_\beta A |m - n|.$$

For positive radii s, t ,

$$\left| \frac{s - t}{1 - st} \right| = \tanh \left(\frac{|2 \operatorname{artanh} s - 2 \operatorname{artanh} t|}{2} \right).$$

Consequently, with $B_\beta = a_\beta A/2$,

$$\rho(w_m^\pm, w_n^\pm) \geq \tanh(B_\beta |m - n|). \quad (1)$$

Since

$$P_\beta := \prod_{j=1}^{\infty} \tanh(B_\beta j) > 0,$$

the product over all other points on the same powered ray is at least P_β^2 , uniformly in the chosen point.

For the opposite ray, let

$$s_0 = r_0^\beta, \quad q_\beta = s_0 |\sin \phi_\beta| > 0.$$

Exactly as for the unpowered rays,

$$\rho(w_m^+, w_n^-) \geq q_\beta \quad (2)$$

and

$$1 - \rho(w_m^+, w_n^-)^2 \leq \frac{(1 - s_m^2)(1 - s_n^2)}{4s_0^2 \sin^2 \phi_\beta}. \quad (3)$$

The function

$$r \mapsto \frac{1 - r^{2\beta}}{1 - r^2}$$

extends continuously and positively to $[r_0, 1]$, with value β at 1. Therefore

$$\sum_{n=0}^{\infty} (1 - s_n^2) \lesssim_{\beta, A} \sum_{n=0}^{\infty} (1 - r_n^2) < \infty.$$

Equations (2)–(3), together with $-\log x \leq (1 - x)/q_\beta$ for $x \in [q_\beta, 1]$, give a positive uniform lower bound for the full opposite-ray product. Combining this with (1) proves that $\{w_n^+, w_n^-\}_{n \geq 0}$ is a Carleson sequence. Taking $\beta = 1$ proves in particular that the original spectrum is Carleson.

Now fix $\alpha > 0$. Suppose first that $\alpha\theta \in \pi\mathbb{Z}$. Then

$$(z_n^+)^{\alpha} = (z_n^-)^{\alpha} \quad (n \in \mathbb{N}_0).$$

It follows that the two corresponding coordinates of $D^{\alpha k}g$ agree for every k . Hence

$$b_n := \delta_n^+ - \delta_n^- \quad \text{satisfies} \quad \langle b_n, D^{\alpha k}g \rangle = 0 \quad (k \in \mathbb{N}_0).$$

The vectors b_n are mutually orthogonal, so the orthogonal complement is infinite-dimensional.

Finally suppose that $\alpha\theta \notin \pi\mathbb{Z}$. By the powered claim, the sequence

$$\mathcal{W}_\alpha = \{w_n^\pm = (z_n^\pm)^\alpha : n \in \mathbb{N}_0\}$$

is Carleson. Let E be diagonal with spectrum $\mathcal{W}_\alpha$ and let $h_n^\pm = \sqrt{1 - r_n^{2\alpha}}$. The canonical Carleson-frame theorem [2, Theorem 1.1] says that $\{E^k h\}_{k \geq 0}$ is a frame. The ratios

$$m_n = \sqrt{\frac{1 - r_n^2}{1 - r_n^{2\alpha}}}$$

are bounded above and below away from zero: their squares are positive and continuous on $[r_0, 1]$, with limiting value $1/\alpha$ at 1. Thus $M\delta_n^\pm = m_n\delta_n^\pm$ defines a bounded invertible diagonal operator. Coordinatewise,

$$D^{\alpha k}g = ME^k h.$$

A bounded invertible image of a frame is a frame, proving the other direction. $\square$

Corollary 2 (Density phase diagram). *For the fixed spectrum in Theorem 1 and every $d > 0$, put*

$$\Lambda_d = \{k/d : k \in \mathbb{N}_0\}.$$

Then $L(\Lambda_d) = d$, and

$$\{D^\lambda g\}_{\lambda \in \Lambda_d} \begin{cases} \text{is a frame,} & d \notin \{\theta/(m\pi) : m \in \mathbb{N}\}, \\ \text{has infinite-dimensional orthogonal complement,} & d = \theta/(m\pi) \text{ for some } m \in \mathbb{N}. \end{cases}$$

Proof. The harmonic-sum calculation gives

$$L(\Lambda_d) = d.$$

Apply Theorem 1 with $\alpha = 1/d$. Resonance means $\theta/d = m\pi$, equivalently $d = \theta/(m\pi)$. $\square$

5 Density-threshold sharpness

Theorem 3. *Let $0 < c < \pi$ and $0 < d \leq c/\pi$. There exist a Carleson sequence $\mathcal{Z} \subset \mathbb{D}_c$ and a countable set $\Lambda \subset [0, \infty)$ such that*

$$L(\Lambda) = d$$

and

$$\dim \left(\overline{\text{span}} \{ D^\lambda g : \lambda \in \Lambda \}^\perp \right) = \infty.$$

Consequently the strict condition $L(\Lambda) > c/\pi$ in [1, Theorem 3.5] is optimal: at equality, the canonical system may fail even to be complete, and may have infinite defect.

Proof. Set

$$\theta = \pi d, \quad \Lambda = \{k/d : k \in \mathbb{N}_0\}.$$

Fix any $A > 0$, put $r_n = \tanh(A(n+1))$, and define

$$z_n^+ = r_n e^{i\theta}, \quad z_n^- = r_n e^{-i\theta} \quad (n \in \mathbb{N}_0).$$

Interlace these two rays to obtain a sequence $\mathcal{Z}$. Since $0 < \theta \leq c < \pi$, the points are distinct and $\mathcal{Z} \subset \mathbb{D}_c$.

Step 1: $\mathcal{Z}$ is a Carleson sequence. For two points on the same ray, the subtraction formula for the hyperbolic tangent gives

$$\rho(z_m^\pm, z_n^\pm) = \frac{|r_m - r_n|}{1 - r_m r_n} = \tanh(A|m - n|). \quad (4)$$

Let

$$P_A = \prod_{k=1}^{\infty} \tanh(Ak).$$

This product is positive because

$$\sum_{k=1}^{\infty} (1 - \tanh(Ak)) < \infty.$$

For a fixed m , the product over the other points on the same ray is

$$\left(\prod_{k=1}^m \tanh(Ak) \right) \left(\prod_{k=1}^{\infty} \tanh(Ak) \right) \geq P_A^2. \quad (5)$$

It remains to control the opposite ray. Write

$$\rho_{mn} = \rho(z_m^+, z_n^-), \quad r_0 = \tanh A, \quad s = \sin \theta > 0.$$

From

$$z_m^+ - z_n^- = (r_m - r_n) \cos \theta + i(r_m + r_n) \sin \theta$$

and $|1 - \overline{z_n^-} z_m^+| \leq 2$, we obtain

$$\rho_{mn} \geq r_0 s =: q > 0. \quad (6)$$

The standard identity for pseudohyperbolic distance gives

$$\begin{aligned} 1 - \rho_{mn}^2 &= \frac{(1 - r_m^2)(1 - r_n^2)}{|1 - r_m r_n e^{-2i\theta}|^2} \\ &\leq \frac{(1 - r_m^2)(1 - r_n^2)}{4r_0^2 s^2}, \end{aligned} \quad (7)$$

because

$$|1 - r_m r_n e^{-2i\theta}|^2 = (1 - r_m r_n)^2 + 4r_m r_n \sin^2 \theta \geq 4r_0^2 s^2.$$

Furthermore,

$$S_A := \sum_{n=0}^{\infty} (1 - r_n^2) = \sum_{n=0}^{\infty} \operatorname{sech}^2(A(n+1)) < \infty.$$

Since $1 - \rho_{mn} \leq 1 - \rho_{mn}^2$, (7) yields, uniformly in m ,

$$\sum_{n=0}^{\infty} (1 - \rho_{mn}) \leq \frac{S_A}{4r_0^2 s^2}.$$

For $x \in [q, 1]$, one has $-\log x \leq (1 - x)/q$. Therefore

$$\prod_{n=0}^{\infty} \rho_{mn} \geq \exp\left(-\frac{S_A}{4qr_0^2 s^2}\right) > 0 \quad (8)$$

uniformly in m . The same estimate applies with the signs reversed. Combining (5) and (8) proves

$$\inf_{z \in \mathcal{Z}} \prod_{\substack{w \in \mathcal{Z} \\ w \neq z}} \rho(z, w) > 0.$$

Thus $\mathcal{Z}$ is a Carleson sequence.

Step 2: $L(\Lambda) = d$. For every fixed $\mu > 1$, the harmonic-sum asymptotic gives

$$\begin{aligned} \lim_{t \rightarrow \infty} \sum_{\lambda \in \Lambda \cap [t, \mu t]} \frac{1}{\lambda} &= \lim_{t \rightarrow \infty} d \sum_{\substack{k \in \mathbb{N} \\ dt \leq k \leq d\mu t}} \frac{1}{k} \\ &= d \log \mu. \end{aligned}$$

After division by $\log \mu$, the expression inside the infimum defining $L(\Lambda)$ equals d for every $\mu > 1$. Hence $L(\Lambda) = d$.

Step 3: infinite defect. Index the coordinate vectors corresponding to $z_n^\pm$ by $\delta_n^\pm$. The two coordinates have the same canonical weight $\sqrt{1 - r_n^2}$. If $\lambda = k/d \in \Lambda$, then the principal-phase definition gives

$$(z_n^\pm)^\lambda = r_n^{k/d} e^{\pm i\theta k/d} = r_n^{k/d} e^{\pm i\pi k} = (-1)^k r_n^{k/d}.$$

Thus the $+$ and $-$ coordinates of $D^\lambda g$ agree for every n and every $\lambda \in \Lambda$. Consequently

$$b_n := \delta_n^+ - \delta_n^- \quad \text{satisfies} \quad \langle b_n, D^\lambda g \rangle = 0 \quad (\lambda \in \Lambda).$$

The nonzero vectors b_n are mutually orthogonal. They span an infinite-dimensional subspace of the orthogonal complement, completing the proof. $\square$

6 Verification, scope, and novelty

Adversarial proof checks

Potential failure	Check
Powering could destroy same-ray separation	The derivative of U_α has a positive infimum, so (1) gives a geometric lower bound for every same-ray factor and hence for the full product.
Powered cross-ray product could collapse	Off resonance the powered rays have distinct boundary points. Equations (2)–(3) give a common factor lower bound and uniformly summable deficits.
Phase wrap could create an unlisted case	The principal phase ϕ_α satisfies $0 < \phi_\alpha < \pi$ exactly when $\alpha\theta \notin \pi\mathbb{Z}$; equality is exactly the collision case.
Frame transfer could change the generator	The ratio $\sqrt{(1 - r_n^2)/(1 - r_n^{2\alpha})}$ extends positively to $r = 1$, so one bounded invertible diagonal operator maps the powered canonical frame to the sampled orbit.
The density calculation ignores $\lambda = 0$	For $t \rightarrow \infty$, the block $[t, \mu t]$ never contains 0.
Only one annihilator is produced	Every radial pair produces a mutually orthogonal vector b_n , giving infinite defect.

Scope

Theorem 1 is a necessary-and-sufficient classification for every arithmetic fractional-power orbit of the stated two-ray spectra. It does not characterize arbitrary spectra or arbitrary nonarithmetic sets Λ , so the broad program stated in the source remains open. Corollary 2 shows that within this model density alone does not determine spanning: at every nonresonant density the system is a frame, while a discrete sequence of resonant densities has infinite defect.

Bounded novelty check

On 23 July 2026, the run indexes were searched for arXiv:2505.04859 and the core phrases. External searches covered the current arXiv v2 page, the published article page, the 2023/2026 source on the mystery of Carleson frames, the July 2026 preprint on block-diagonal Carleson frames, and exact or close combinations of *Carleson systems*, *powered spectrum*, *two rays*, *fractional powers*, *phase collision*, *logarithmic block density*, and c/π . No explicit statement of Theorem 1 was found. The block-diagonal preprint’s redundancy classification is for nonnegative spectra and does not contain this complex two-ray phase diagram.

Novelty confidence is moderate, not high. Remark 2.6 of [1] already observes that if two spectral points have the same N -th power, the analysis matrix for $\{D^{Nk}g\}$ has proportional columns and the system cannot be a frame. The result here parameterizes that mechanism for every real arithmetic step, proves the complementary nonresonant cases are frames, and obtains the exact density phase diagram. The positive direction requires the new powered-spectrum Carleson estimate and bounded diagonal transfer. A human should still judge whether the classification is best presented as a new proposition or as a developed but unstated consequence of the source mechanisms.

Human-review recommendation

Send to a human reviewer. The main checks are the positive lower bound for U'_α , the powered cross-ray product estimate, and the diagonal frame transfer. If accepted, the frame-versus-infinite-defect dichotomy is a complete theorem for the two-ray model and is suitable for a short note; the unrestricted spanning problem must remain labelled open.

References

- [1] I. Krishtal and B. Miller, *Demystifying Carleson Frames*, Applied and Computational Harmonic Analysis (2025), article 101811; arXiv:2505.04859v2.
- [2] O. Christensen, M. Hasannasab, F. M. Philipp, and D. Stoeva, *The mystery of Carleson frames*, Applied and Computational Harmonic Analysis **72** (2024), article 101659; arXiv:2312.11978.